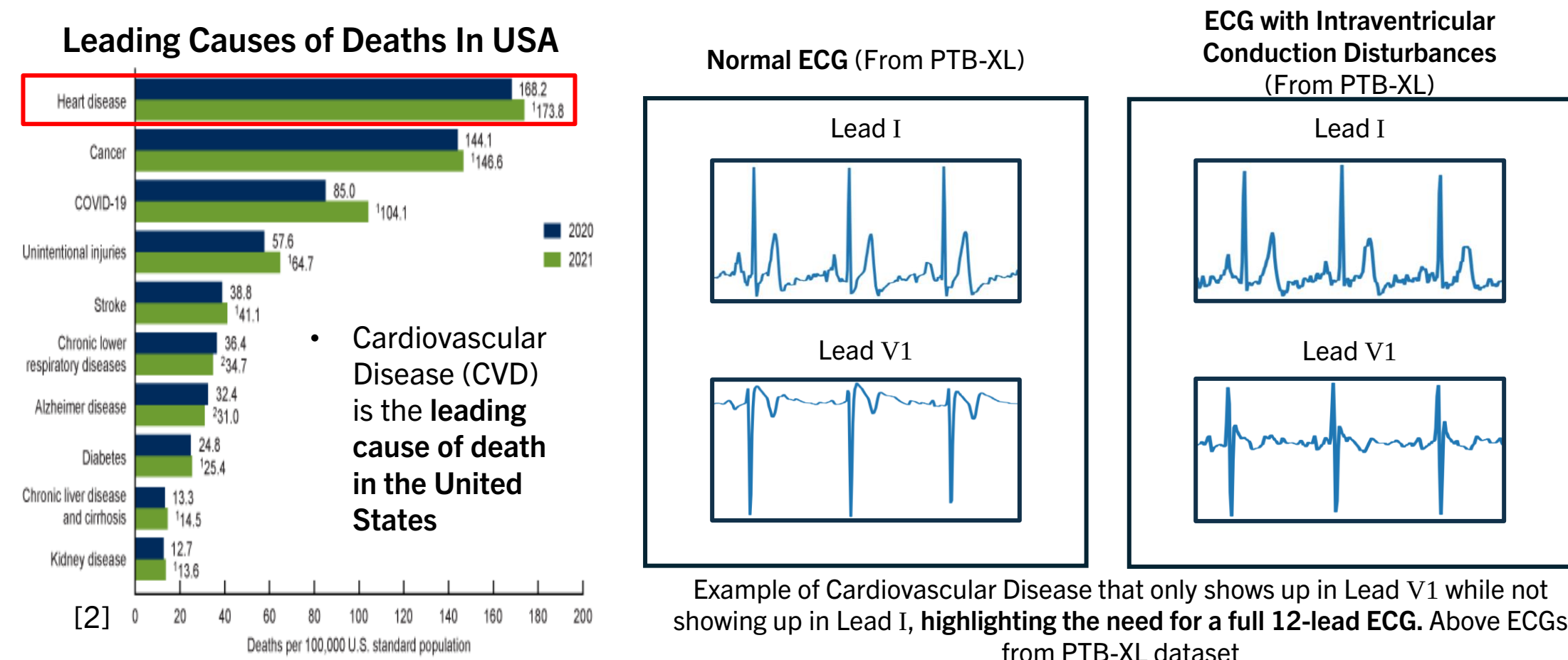
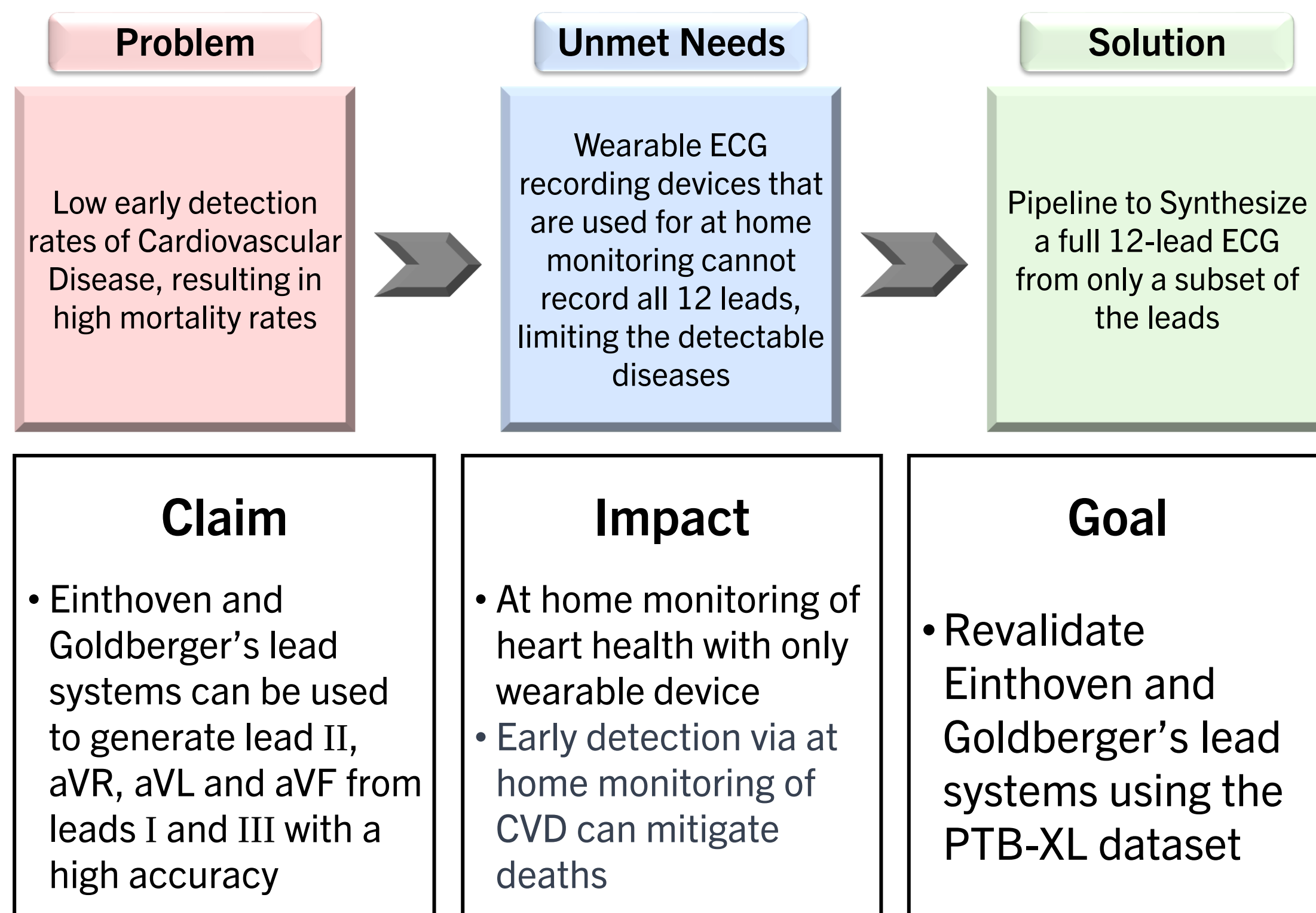


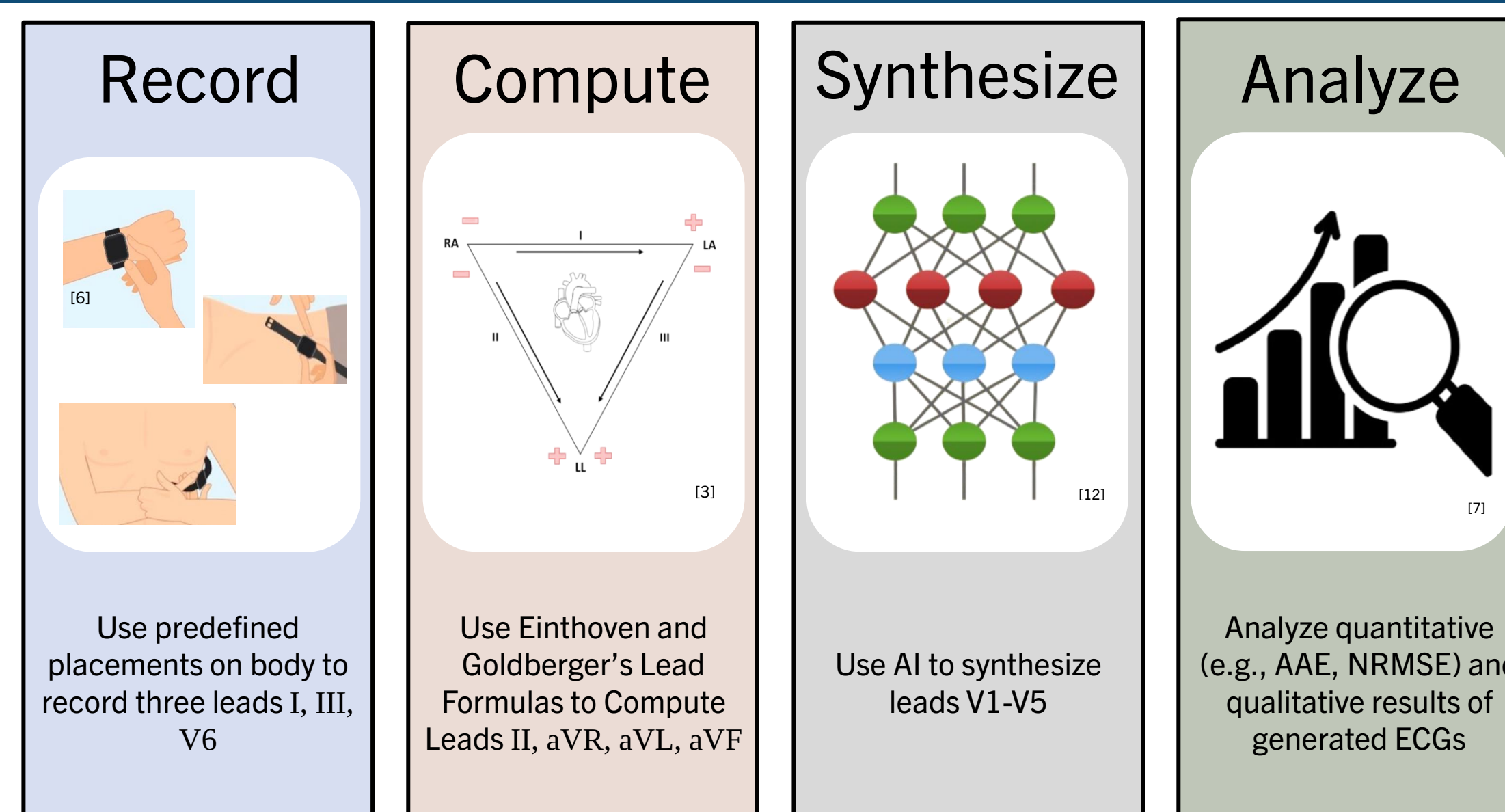
Introduction



- At home monitoring is crucial for detecting Cardiovascular Disease that can show up only weeks in advance
 - Cardiovascular Disease (CVD) can show up only **weeks** in advance
- Currently **ONLY Atrial Fibrillation and Atrial Flutter** can be detected from a standard ECG recorded from a wearable device
 - Ventricular diseases and Pericarditis **CANNOT** be detected
- Asymptomatic individuals with CVD rarely get a full 12-lead ECG taken, even at annual check-ups
- Access to 12-Lead ECGs in low and middle-income countries could be as **low as 10%**

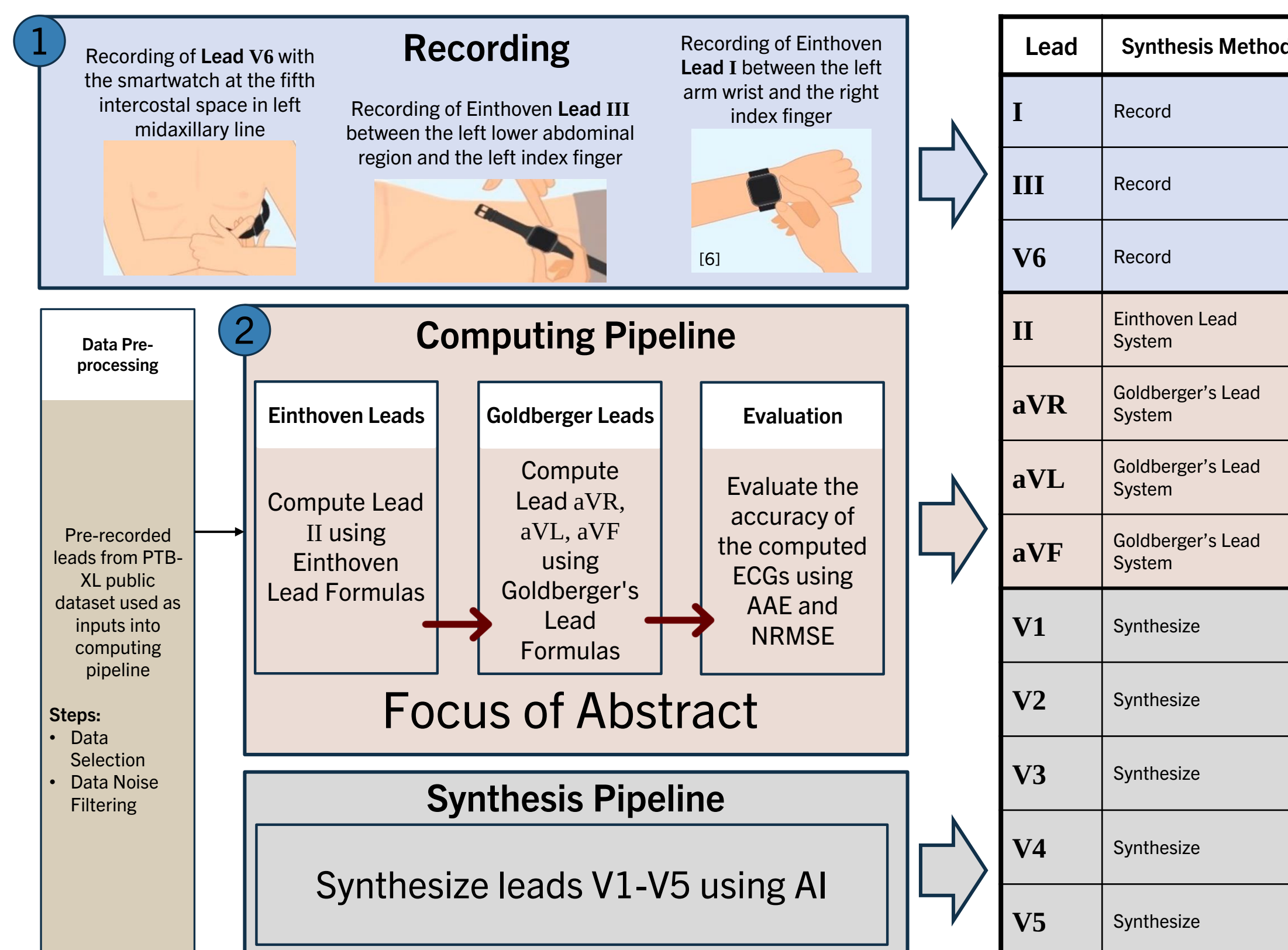


4-Step Workflow



Step 1 → Step 2 → Step 3 → Step 4

Strategy



Lead	Synthesis Method
I	Record
III	Record
V6	Record
II	Einthoven Lead System
aVR	Goldberger's Lead System
aVL	Goldberger's Lead System
aVF	Goldberger's Lead System
V1	Synthesize
V2	Synthesize
V3	Synthesize
V4	Synthesize
V5	Synthesize

Comprehensive Validation of Einthoven's and Goldberger's Electrocardiographic Lead Systems Using the PTB-XL Dataset



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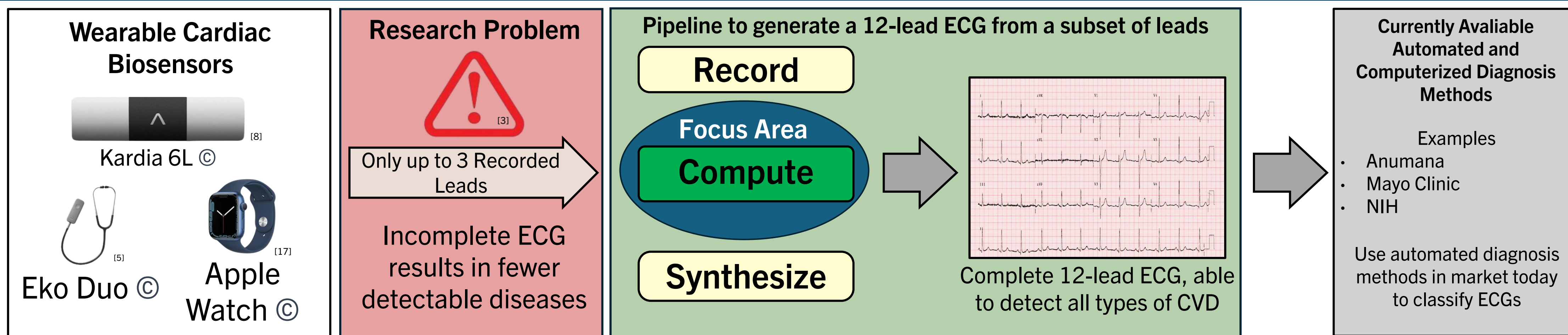
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Research Focus



Cardiovascular disease (CVD) is the leading cause of death in the United States. Early detection and at home monitoring can prevent a lot of the associated fatalities. Smart wearable devices are ideal for at-home monitoring but only record three leads, limiting detectable diseases. This project aims to create a pipeline to generate a diagnostic-quality 12-lead ECG, necessary for comprehensive CVD detection.

One of the three major parts of the 12-lead generation from a subset of leads is computation of leads II, aVR, aVF, and aVL. Einthoven and Goldberger's lead formulae are the best and only methods for computing these leads but to ensure that the whole 12-lead ECG is diagnostic quality both Einthoven and Goldberger's lead formulae must be revalidated with the PTB-XL dataset.

Results

Lead	Average Absolute Error (AAE) – lower is better	Root Mean Squared Error (RMSE) – lower is better	Normalized Root Mean Squared Error (NRMSE) – lower is better
Lead II	0.0002597 mV	0.0004899 mv	0.00053655
Lead aVR	0.0003348 mV	0.00045634 mv	0.0005191
Lead aVL	0.0003343 mV	0.00044721 mv	0.00064288
Lead aVF	0.0003334 mV	0.00044917 mv	0.00062393

Error Metrics

Average Absolute Error (AAE)

The Average Absolute Error (AAE) metric measures the difference between values computed and actual (true values) by computing the average of the absolute differences across all data points. It provides an, assessment of the average magnitude of errors in prediction or estimation, without distinguishing between overestimations and underestimations.

$$AAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

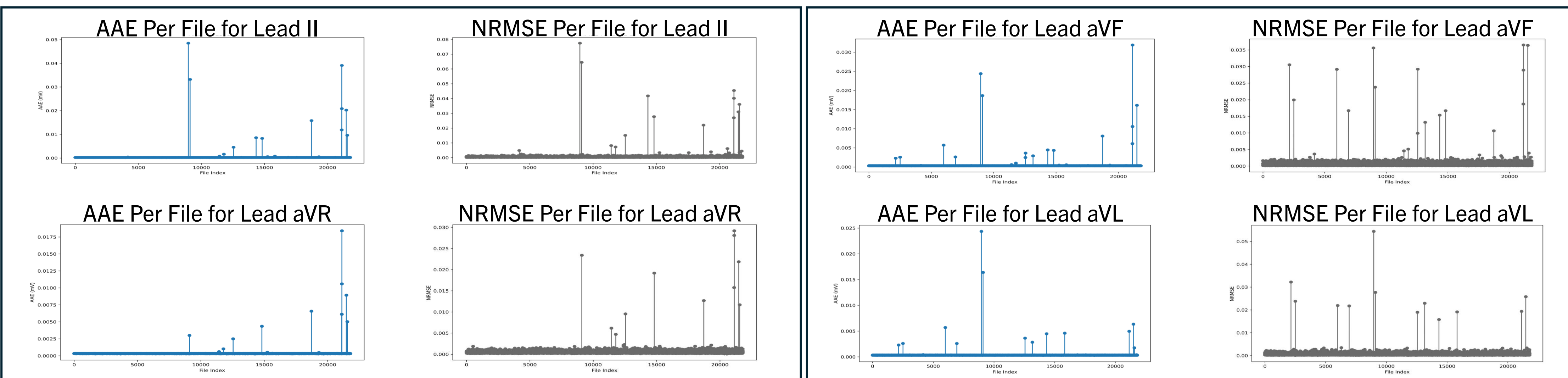
Root Mean Squared Error (RMSE)

The Root Mean Squared Error (RMSE) quantifies the difference between values predicted by a model and the actual values observed. It is calculated by taking the square root of the average of the squared differences between predicted and actual values. This squaring process penalizes larger errors more severely than smaller ones, making RMSE particularly sensitive to large deviations in prediction accuracy.

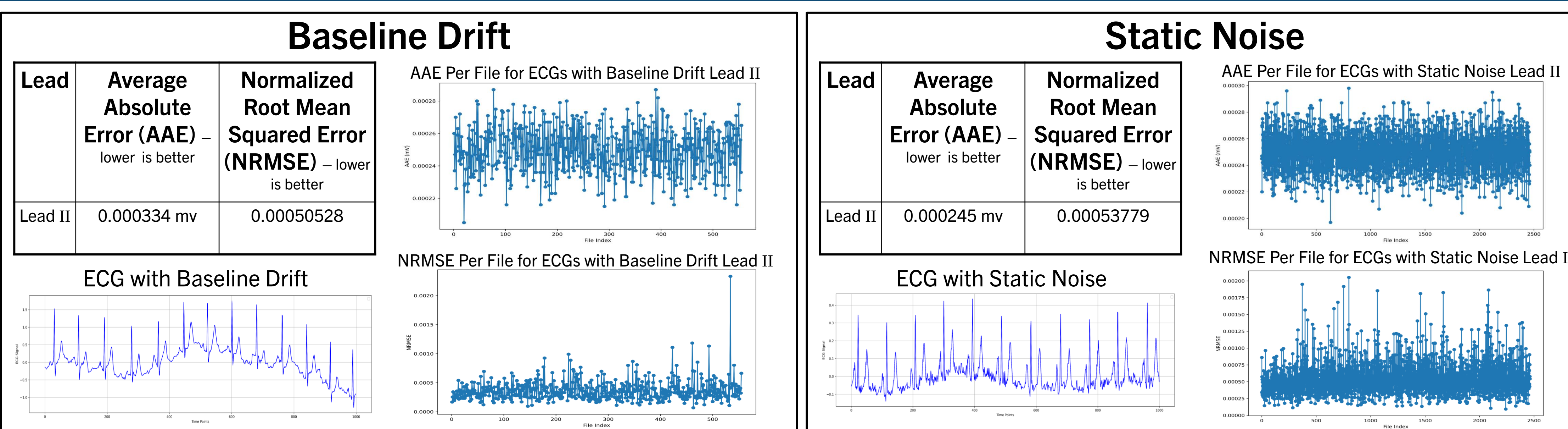
$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Normalized Root Mean Squared Error (NRMSE)

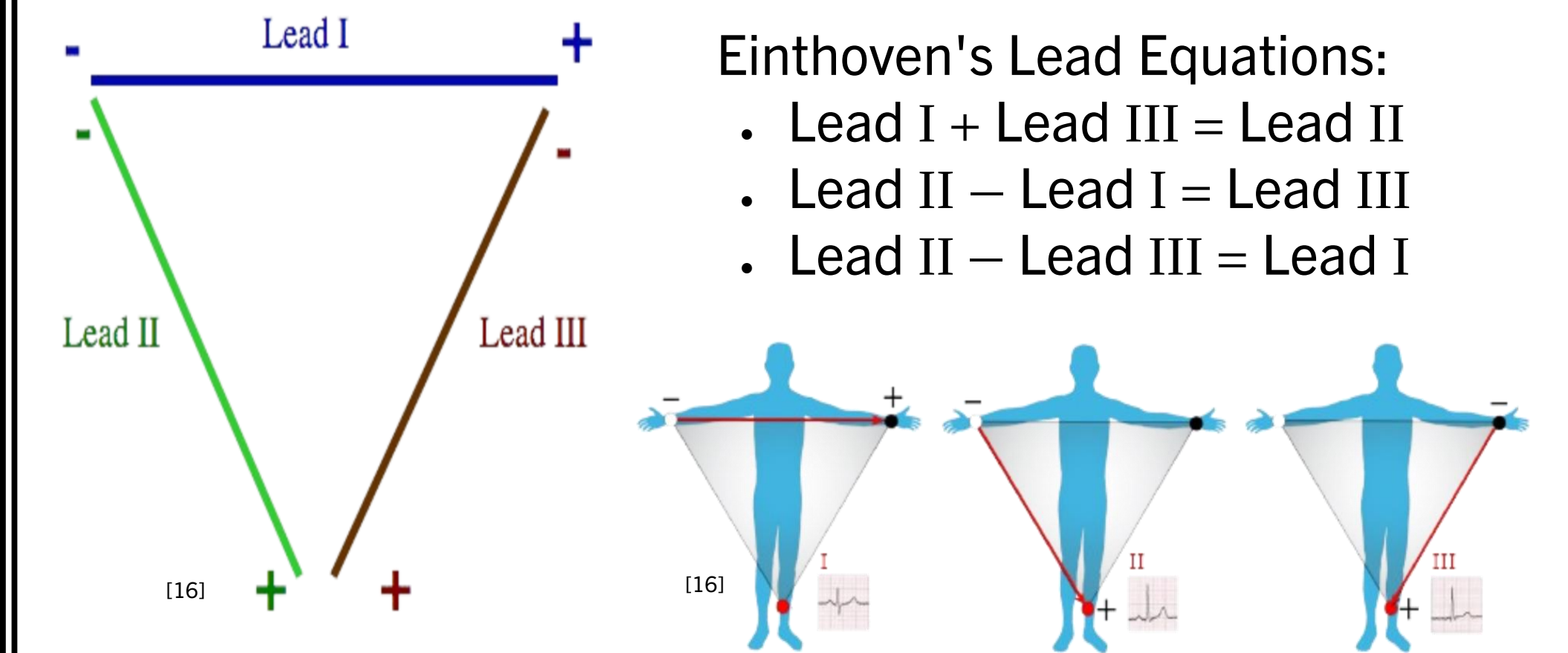
The Normalized Root Mean Squared Error (NRMSE) adjusts the Root Mean Squared Error (RMSE) by normalizing between 0 and 1. This makes NRMSE a scale-invariant metric that allows for comparisons across different datasets. It is useful for evaluating accuracy relative to the variability in the data, providing a consistent measure of prediction error.

$$NRMSE = \frac{RMSE}{\text{range}(y)}$$


Noise Analysis

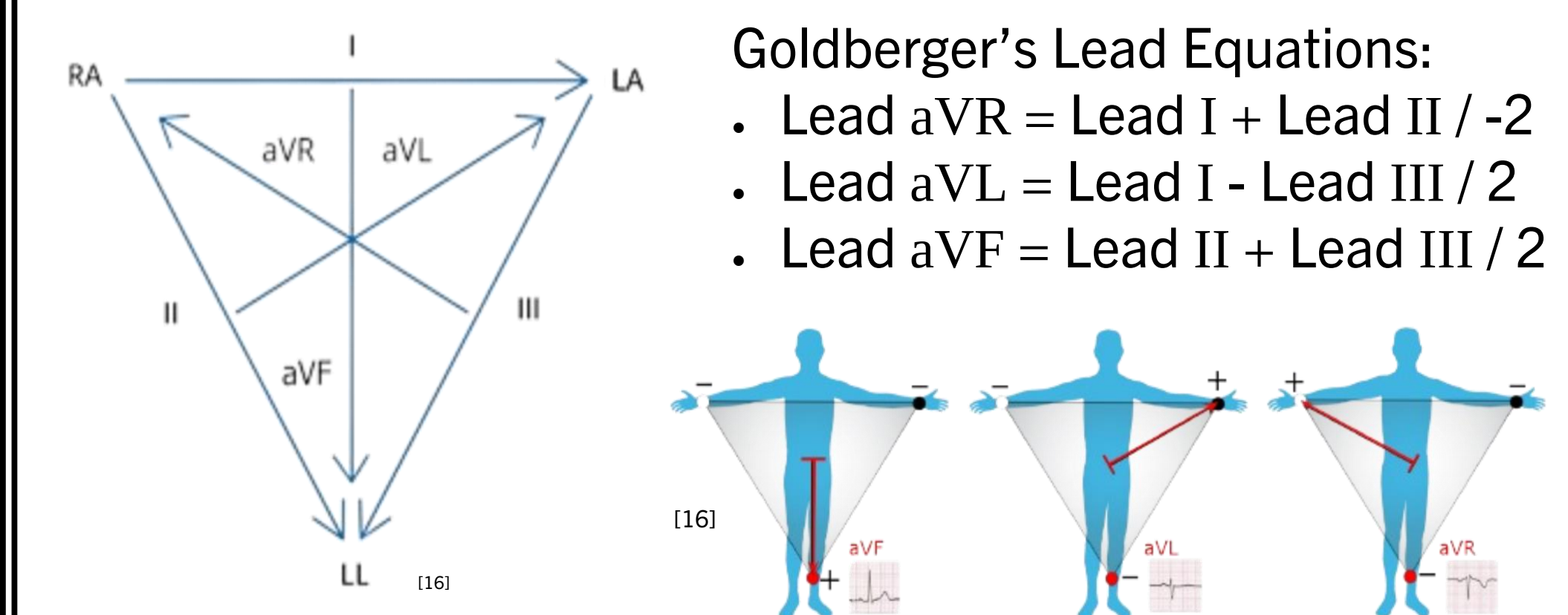


Einthoven Lead System



- Einthoven's lead formulas are a set equations used in electrocardiography to derive the electrical activity of the heart from different angles. These formulas involve Lead I, Lead II, and Lead III, which represent different viewpoints of the heart's electrical axis in the frontal plane.

Goldberger's Lead System



- Goldberger's lead formulas are a set formulae used to derive the unipolar limb leads (aVR, aVL, and aVF) from the bipolar limb leads (I, II, and III). The unipolar limb leads are known as redundant leads and provide a secondary view of the frontal plane of the heart.

Conclusions

Highlights

- Revalidated of the Einthoven lead systems to compute lead II from leads I and III, as evidenced by the low error values from AAE, NRMSE and RMSE
- The revalidation of Goldberger's lead systems to compute lead aVR, aVL and aVF using AAE, RMSE, and NRMSE. Notably low error values demonstrated their high accuracy and reliability
- Einthoven and Goldberger's lead formulae exhibit similar AAE and NRMSE values when applied to ECGs affected by baseline drift and static noise, proving their reliability across various noise conditions.

Future Work

- Develop an generative AI method to synthesize leads V1-V5

Impact

- Allows advanced CVD detection using wearable devices** – wearable devices can be used to generate complete 12-lead diagnostic quality ECGs
- Increases early detection rates of cardiovascular disease** – enables at home monitoring of CVD
- Demonstrates effectiveness of wearables in healthcare** - novel pipeline can be applied for the generation of other electrophysiological signals (e.g. EEGs)

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